

EE3621: Digital Signal Processing

Lecture 19: IIR Filter — Parallel Form Realization

(III B.Tech EEE)

Comprehensive Lecture Notes & Faculty Textbook Reference

Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 06:00 (6 mins):** Motivation for Parallel Realization: From multiplicative decomposition (Cascade: $H(z) = \prod H_k(z)$) to additive decomposition (Parallel: $H(z) = C + \sum H_k(z)$). Concurrency, throughput, and noise decoupling advantages.
 - **06:00 – 16:00 (10 mins):** Mathematical Foundations of Partial Fraction Expansion (PFE): Handling proper ($M < N$) vs. improper ($M \geq N$) rational functions, polynomial long division for direct feedthrough C , decomposition directly in z^{-1} , and algebraic grouping of complex conjugate pole pairs into real second-order sections (SOS).
 - **16:00 – 25:00 (9 mins):** Canonical Parallel Biquad Circuit Architecture: State update equations, branch output equations, and proving why parallel biquads have numerator degree ≤ 1 (saving 1 multiplier per section compared to Cascade biquads).
 - **25:00 – 32:00 (7 mins):** Parallel Form I vs. Parallel Form II & Finite Word-Length Superiority: Total decoupling of roundoff quantization noise (zero inter-stage noise amplification) and low-latency adder tree pipelining.
 - **32:00 – 38:00 (6 mins):** Step-by-Step Worked Numerical Examples: Improper 2nd-order system ($M = N = 2$ with direct term $C = 8$), 4th-order mixed-pole system, and reverse synthesis from parallel SFG.
 - **38:00 – 40:00 (2 mins):** Faculty Insights, Common Misconceptions, and University Semester Exam Traps.
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1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Formulate** the additive partial fraction expansion $H(z) = C + \sum_{k=1}^K H_k(z)$ for both strictly proper ($M < N$) and improper ($M \geq N$) rational transfer functions.
2. **Extract** the direct feedthrough scalar coefficient $C = b_N/a_N$ (or polynomial quotient) using polynomial long division.
3. **Combine** complex conjugate residue pairs algebraically to produce second-order sections with strictly real coefficients and numerator degree ≤ 1 .

4. **Synthesize** canonical Direct Form II parallel biquads with minimum memory registers ($\max(M_k, N_k) \leq 2$).
 5. **Audit** exact hardware resource requirements (multipliers, adders, delay registers) and contrast them with Direct Form II and Cascade Form.
 6. **Explain** the critical finite word-length advantages of parallel realization (complete noise decoupling and suppression of limit-cycle oscillations).
 7. **Differentiate** between Parallel Form I (expansion in z^{-1}) and Parallel Form II (expansion in z via $H(z)/z$).
 8. **Evaluate** impulse responses and difference equations of parallel structures step-by-step.
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2. MATHEMATICAL FOUNDATIONS OF PARALLEL REALIZATION

2.1 From Product to Sum: The Additive Decomposition Paradigm

In Lecture 18, we decomposed high-order IIR transfer functions multiplicatively into a cascade of modular sub-systems:

$$H(z) = g \prod_{k=1}^K H_k(z) \iff h[n] = g \cdot h_1[n] * h_2[n] * \dots * h_K[n] \quad (1)$$

While the Cascade form successfully solves the root sensitivity catastrophe of high-order Direct Forms, it processes data in a strictly **sequential pipeline**: the output of Section 1 must be calculated before Section 2 can begin, and roundoff noise generated in early sections is progressively filtered and amplified by all downstream sections.

The **Parallel Form Realization** takes an entirely different, additive approach. By utilizing the linearity property of the Z -transform:

$$\mathcal{Z}^{-1} \left\{ \sum_{k=1}^K H_k(z) \right\} = \sum_{k=1}^K h_k[n] \quad (2)$$

we decompose the rational transfer function into an **additive sum of independent, low-order sub-systems** operating in parallel:

$$H(z) = C + \sum_{k=1}^K H_k(z) \iff y[n] = Cx[n] + \sum_{k=1}^K y_k[n] \quad (3)$$

The input sequence $x[n]$ is broadcast simultaneously to all branches, each branch computes its output $y_k[n]$ concurrently, and their individual outputs are summed at a final accumulation junction to yield $y[n]$.

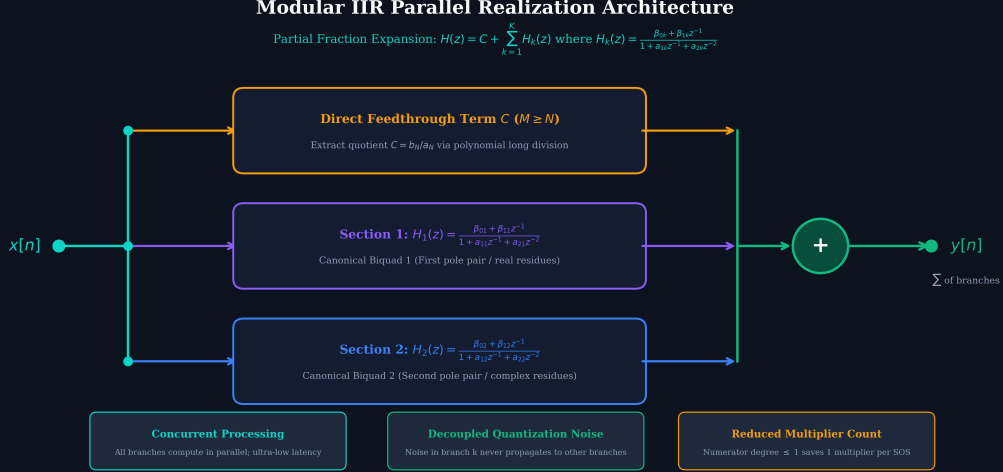


Figure 1: Modular IIR Parallel Realization Architecture. The input $x[n]$ is distributed concurrently to a direct feedthrough branch C and K parallel second-order sections $H_k(z)$. The branch outputs are summed at the final accumulation junction to produce $y[n]$.

2.2 Handling Proper vs. Improper Transfer Functions ($M \geq N$ vs $M < N$)

Consider a general rational IIR transfer function:

$$H(z) = \frac{B(z)}{A(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}} \quad (4)$$

Partial fraction expansion cannot be applied directly if the degree of the numerator in z^{-1} is greater than or equal to the degree of the denominator ($M \geq N$). We must distinguish two cases:

2.2.1 Case 1: Strictly Proper System ($M < N$)

When the numerator degree is strictly less than the denominator degree ($M < N$), the transfer function decays to zero as $z^{-1} \rightarrow \infty$ (i.e., at $z = 0$). No direct polynomial division is needed, and the direct feedthrough constant is identically zero:

$$C = 0 \quad (5)$$

2.2.2 Case 2: Improper System ($M = N$)

In the vast majority of practical digital filters (such as Butterworth, Chebyshev, and Elliptic filters designed via bilinear transformation), the numerator and denominator have equal degrees: $M = N$. Before partial fraction expansion can be performed, we must perform **polynomial long division** in powers of z^{-1} to extract the quotient C :

$$H(z) = C + \frac{B_1(z)}{A(z)} \quad (6)$$

where $B_1(z)$ has degree at most $N - 1$. The direct constant C is found by taking the limit as $z^{-1} \rightarrow \infty$:

$$C = \lim_{z^{-1} \rightarrow \infty} H(z) = \frac{b_N z^{-N}}{a_N z^{-N}} = \frac{b_N}{a_N} \quad (7)$$

Alternatively, evaluating at $z \rightarrow \infty$ ($z^{-1} \rightarrow 0$):

$$H(z) \Big|_{z^{-1}=0} = \frac{b_0}{a_0} = b_0 = C + \sum_{k=1}^K H_k(z) \Big|_{z^{-1}=0} \quad (8)$$

In the time domain, C represents an **instantaneous direct feedthrough** from input to output: $Cx[n]$.

2.2.3 Case 3: Highly Improper System ($M > N$)

If $M > N$, polynomial long division yields an FIR polynomial quotient of degree $M - N$:

$$H(z) = \sum_{m=0}^{M-N} c_m z^{-m} + \sum_{k=1}^K H_k(z) \quad (9)$$

This is realized as an FIR tapped delay line in parallel with the IIR recursive branches.

2.3 2.3 Partial Fraction Expansion in Powers of z^{-1}

Assuming distinct poles, the strictly proper rational remainder $B_1(z)/A(z)$ is expanded into first-order partial fractions in powers of z^{-1} :

$$H(z) = C + \sum_{i=1}^N \frac{R_i}{1 - p_i z^{-1}} \quad (10)$$

where the residue R_i corresponding to pole p_i is evaluated using Heaviside's residue formula:

$$R_i = (1 - p_i z^{-1}) H(z) \Big|_{z^{-1}=1/p_i} \quad (11)$$

2.4 2.4 Combining Complex Conjugate Pole Pairs into Real Biquads

Physical hardware must perform arithmetic using real numbers. For real-coefficient filters, any complex pole $p_k = r_k e^{j\theta_k}$ always occurs alongside its complex conjugate $p_k^* = r_k e^{-j\theta_k}$. Their corresponding residues are also complex conjugates:

$$R_k = |R_k| e^{j\phi_k}, \quad R_k^* = |R_k| e^{-j\phi_k} \quad (12)$$

We combine each complex conjugate pair into a single second-order section (SOS):

$$\begin{aligned} H_k(z) &= \frac{R_k}{1 - p_k z^{-1}} + \frac{R_k^*}{1 - p_k^* z^{-1}} \\ &= \frac{R_k(1 - p_k^* z^{-1}) + R_k^*(1 - p_k z^{-1})}{(1 - p_k z^{-1})(1 - p_k^* z^{-1})} \\ &= \frac{(R_k + R_k^*) - (R_k p_k^* + R_k^* p_k) z^{-1}}{1 - (p_k + p_k^*) z^{-1} + (p_k p_k^*) z^{-2}} \end{aligned} \quad (13)$$

Using Euler's identity:

- $R_k + R_k^* = 2|R_k| \cos(\phi_k) = 2\text{Re}(\mathbf{R}_k)$
- $R_k p_k^* + R_k^* p_k = 2\text{Re}(R_k p_k^*) = 2\mathbf{r}_k |\mathbf{R}_k| \cos(\phi_k - \theta_k)$
- $p_k + p_k^* = 2r_k \cos(\theta_k)$
- $p_k p_k^* = |p_k|^2 = r_k^2$

This yields the canonical second-order parallel section:

$$\mathbf{H}_k(z) = \frac{\beta_{0k} + \beta_{1k}z^{-1}}{1 + \mathbf{a}_{1k}z^{-1} + \mathbf{a}_{2k}z^{-2}} \quad (14)$$

where all coefficients are **strictly real**:

$$\beta_{0k} = 2\text{Re}(R_k) \quad (15)$$

$$\beta_{1k} = -2\text{Re}(R_k p_k^*) \quad (16)$$

$$a_{1k} = -2r_k \cos(\theta_k) \quad (17)$$

$$a_{2k} = r_k^2 \quad (18)$$

The Fundamental Architectural Theorem of Parallel Biquads: Notice that the numerator in (14) has degree **at most 1** ($\beta_{0k} + \beta_{1k}z^{-1}$). There is **no** z^{-2} **term** in the numerator! Consequently, each parallel second-order section requires only **2 feedforward multipliers** (β_{0k} and β_{1k}), compared to **3 feedforward multipliers** (b_{k0}, b_{k1}, b_{k2}) in a Cascade second-order section. Over K biquads, the Parallel form saves K hardware multipliers!

3. CANONICAL PARALLEL BIQUAD ARCHITECTURE (DIRECT FORM II SOS)

Each second-order branch $H_k(z)$ in the parallel realization is implemented using the canonical Direct Form II structure to minimize delay registers.

3.1 Signal Flow Graph and Difference Equations

The input to the k -th branch is the global filter input $x[n]$. Defining the intermediate state variable sequence $W_k(z)$:

$$W_k(z) = \frac{X(z)}{1 + a_{1k}z^{-1} + a_{2k}z^{-2}} \quad (19)$$

and the branch output:

$$Y_k(z) = (\beta_{0k} + \beta_{1k}z^{-1})W_k(z) \quad (20)$$

Taking the inverse Z -transform yields the two governing time-domain equations:

$$\mathbf{w}_k[\mathbf{n}] = \mathbf{x}[\mathbf{n}] - \mathbf{a}_{1k}\mathbf{w}_k[\mathbf{n} - 1] - \mathbf{a}_{2k}\mathbf{w}_k[\mathbf{n} - 2] \quad (\text{Branch State Equation}) \quad (21)$$

$$\mathbf{y}_k[\mathbf{n}] = \beta_{0k}\mathbf{w}_k[\mathbf{n}] + \beta_{1k}\mathbf{w}_k[\mathbf{n} - 1] \quad (\text{Branch Output Equation}) \quad (22)$$

The overall system output is obtained by summing all branch outputs along with the direct feedthrough path:

$$\mathbf{y}[\mathbf{n}] = \mathbf{C}\mathbf{x}[\mathbf{n}] + \sum_{k=1}^K \mathbf{y}_k[\mathbf{n}] \quad (\text{Total System Output}) \quad (23)$$

Canonical Direct Form II Parallel Second-Order Section (SOS / Biquad)

$$\text{Transfer Function: } H_k(z) = \frac{\beta_{0k} + \beta_{1k}z^{-1}}{1 + a_{1k}z^{-1} + a_{2k}z^{-2}} \quad (\text{Numerator Degree} \leq 1)$$

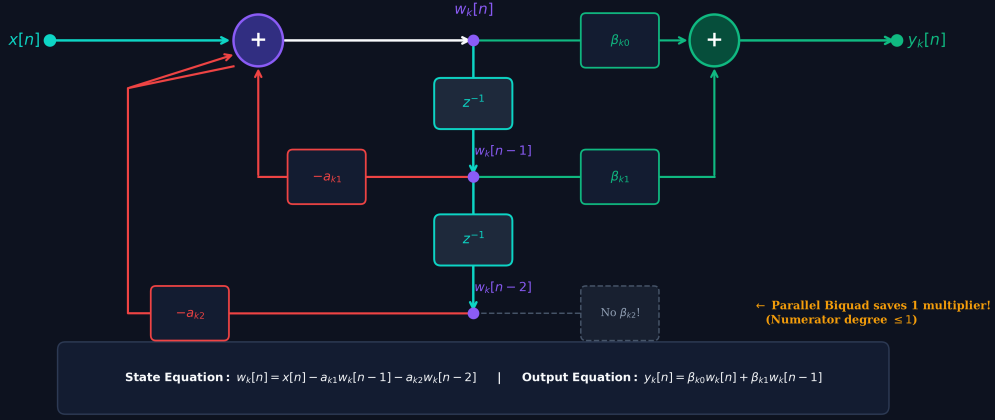


Figure 2: Canonical Direct Form II Parallel Second-Order Section (SOS / Biquad). The shared 2-register delay spine (z^{-1}, z^{-1}) provides states for feedback ($-a_{1k}, -a_{2k}$) and feedforward taps (β_{0k}, β_{1k}). Note the complete absence of a $\beta_{2k}z^{-2}$ tap, saving one multiplier per section.

3.2 First-Order Real Pole Sections

For a real pole $p_k \in \mathbb{R}$ with real residue $A_k \in \mathbb{R}$:

$$H_k(z) = \frac{A_k}{1 - p_k z^{-1}} \quad (24)$$

Its canonical realization requires:

- Exactly 1 delay register ($w_k[n-1]$).
- 2 multipliers: one feedback multiplier ($p_k = -a_{1k}$) and one feedforward multiplier (A_k).
- 1 adder for the feedback state equation: $w_k[n] = x[n] + p_k w_k[n-1]$, with output $y_k[n] = A_k w_k[n]$.

4. PARALLEL FORM I VS. PARALLEL FORM II & FINITE WORD-LENGTH ADVANTAGES

4.1 Parallel Form I vs. Parallel Form II (Oppenheim & Schaffer Formulation)

In classic digital signal processing literature, two conventions exist for formulating the partial fraction expansion:

- **Parallel Form I (Expansion directly in z^{-1}):**

$$H(z) = C + \sum_{k=1}^K \frac{\beta_{0k} + \beta_{1k}z^{-1}}{1 + a_{1k}z^{-1} + a_{2k}z^{-2}} \quad (25)$$

This is the standard modern engineering formulation. The transfer function is treated as a rational function of z^{-1} , directly matching the causal delay elements (z^{-1}) of digital hardware.

- **Parallel Form II (Expansion in z via $H(z)/z$):** In older textbooks, $H(z)$ is expressed in positive powers of z , $H(z)/z$ is expanded into partial fractions, and the result is multiplied back by z :

$$\frac{H(z)}{z} = \frac{c_0}{z} + \sum_{k=1}^K \frac{\gamma_k}{z - p_k} \implies H(z) = c_0 + \sum_{k=1}^K \frac{\gamma_k}{1 - p_k z^{-1}} \quad (26)$$

While mathematically equivalent, expanding directly in z^{-1} (Parallel Form I) is far less prone to algebraic indexing errors during examination calculations.

4.2 4.2 Total Decoupling of Roundoff Quantization Noise

One of the most profound practical engineering advantages of the Parallel Form over the Cascade Form is its **complete immunity to inter-stage noise propagation**:

- **In Cascade Realization:** Each stage k introduces arithmetic roundoff noise $e_k[n]$ due to multiplier quantization. Because the stages are connected in series, the noise $e_1[n]$ produced in Stage 1 is filtered and amplified by all downstream transfer functions $H_2(z), H_3(z), \dots, H_K(z)$:

$$\sigma_{y,\text{cascade}}^2 = \sigma_e^2 \sum_{k=1}^K \frac{1}{2\pi} \int_{-\pi}^{\pi} \left| \prod_{m=k+1}^K H_m(e^{j\omega}) \right|^2 d\omega \quad (27)$$

If downstream stages have high- Q resonant poles, the roundoff noise is magnified catastrophically.

- **In Parallel Realization:** The sections operate independently side-by-side. The roundoff noise $e_k[n]$ generated inside Branch k travels **directly to the output adder** without passing through any other filter sections:

$$\sigma_{y,\text{parallel}}^2 = K \cdot \sigma_e^2 \quad (28)$$

There is **zero inter-section noise amplification!** This makes the Parallel Form the structure of choice for fixed-point audio processing and medical instrumentation where extreme signal purity is required.

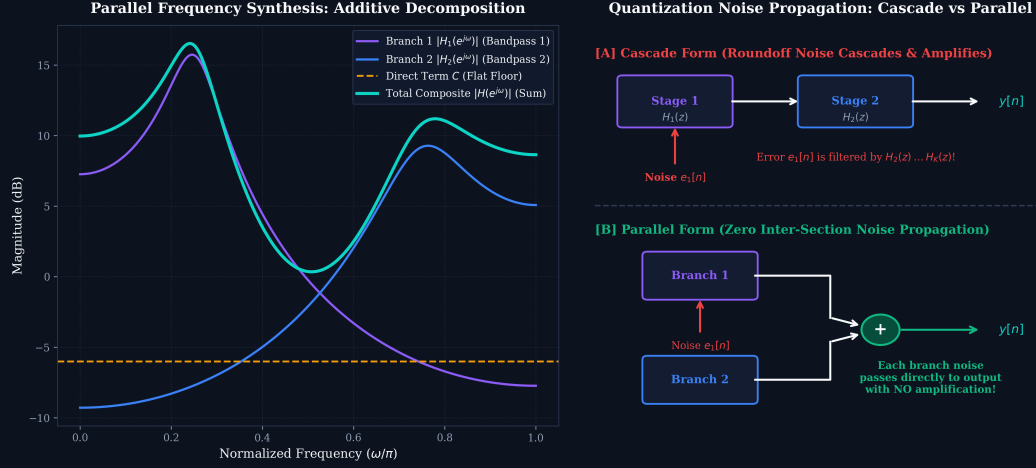


Figure 3: Parallel vs. Cascade Architectural Analysis. (Left) Parallel frequency synthesis: individual bandpass branch responses $|H_1(e^{j\omega})|$ and $|H_2(e^{j\omega})|$ sum vectorially to form the composite frequency response with the direct feedthrough floor C . (Right) Quantization noise model: Cascade form amplifies Stage 1 noise across all downstream sections, while Parallel form completely decouples noise sources.

4.3 Concurrency, Latency, and No Pole-Zero Pairing Dilemma

1. **Concurrency and Ultra-Low Latency:** Because all branches receive $x[n]$ simultaneously, an FPGA or parallel DSP processor can execute all K biquad state equations concurrently. The critical path latency is only:

$$T_{\text{latency}} = T_{\text{mult}} + 2T_{\text{add}} + \lceil \log_2(K + 1) \rceil T_{\text{add}} \quad (29)$$

In contrast, Cascade form has a critical path latency that scales linearly with order: $O(K)$.

2. **No Pole-Zero Pairing Dilemma:** In Cascade form, engineers must solve a complex combinatorial optimization problem to pair poles with zeros ($\theta_z \approx \theta_p$) and order sections. In Parallel form, the zeros are generated implicitly by the additive interference of the parallel branches. **No pole-zero pairing or section ordering is required!**

5. 5. HARDWARE & COMPUTATIONAL COMPLEXITY AUDIT

Table 1 presents a rigorous comparative audit across all four fundamental IIR realization structures for an N -th order filter (assuming $M = N$ even, $K = N/2$).

Architectural Metric	Direct Form I	Direct Form II	Cascade (DF-II SOS)	Parallel (DF-II SOS)
Canonical Status	Non-Canonical	Canonical	Canonical	Canonical
Total Delay Registers (z^{-1})	$2N$	N	N	N
Total Multipliers	$2N + 1$	$2N + 1$	$2N + 1$	$2N + 1$ ($2K$ ff + $2K$ fb + C)
Total Adders	$2N$	$2N$	$2N$	$2N$ ($2K$ biquad + K bus)
Critical Path Latency	1 Mult + $2N$ Adds	1 Mult + $2N$ Adds	K Mults + $2K$ Adds	1 Mult + $(2 + \log_2 K)$ Adds
Root Sensitivity	Catastrophic ($O(N!)$)	Catastrophic ($O(N!)$)	Extremely Low	Extremely Low
Inter-Section Noise Amplification	N/A	N/A	High (Cascaded)	Zero (Decoupled)
Pole-Zero Pairing Required?	No	No	Yes (Combinatorial)	No (Additive Synthesis)
Independent Zero Tuning?	Yes	Yes	Yes (per biquad)	No (Interference Zeros)

Table 1: Comprehensive Architectural Complexity and Performance Comparison.

The Primary Trade-off between Cascade and Parallel Forms:

- **Choose Cascade Form** when transmission zeros must be placed precisely on the unit circle (e.g., notch filters, elliptic filters with exact stopband nulls), because Cascade biquad numerator coefficients directly define zero locations.
- **Choose Parallel Form** when low roundoff noise, high pipelined throughput, and concurrent FPGA processing are the primary design drivers.

6. STEP-BY-STEP WORKED NUMERICAL EXAMPLES

6.1 Example 19.1: Improper Second-Order System ($M = N = 2$ with Direct Term C)

Problem Statement: An IIR digital filter is described by the transfer function:

$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - 0.75z^{-1} + 0.125z^{-2}} \quad (30)$$

- Determine whether the transfer function is strictly proper or improper.
- Perform polynomial long division to extract the direct feedthrough term C .
- Find the poles and perform partial fraction expansion in powers of z^{-1} .
- Formulate the time-domain difference equations for all branches and the total output.
- Construct the complete hardware signal flow graph.
- Compute the impulse response samples $y[0], y[1], y[2]$ step-by-step.

Solution:

(a) Properness Classification: Numerator degree $M = 2$, denominator degree $N = 2$. Since $M = N$, the system is ****improper****. A non-zero direct feedthrough constant C must be extracted before partial fraction expansion.

(b) Extraction of Direct Term C : Using the asymptotic formula (7):

$$C = \frac{b_2}{a_2} = \frac{1}{0.125} = 8 \quad (31)$$

Alternatively, using synthetic polynomial long division in powers of z^{-1} :

$$(1 + 2z^{-1} + z^{-2}) = 8 \cdot (1 - 0.75z^{-1} + 0.125z^{-2}) + R(z^{-1}) \quad (32)$$

Expanding:

$$8 \cdot (1 - 0.75z^{-1} + 0.125z^{-2}) = 8 - 6z^{-1} + z^{-2} \quad (33)$$

$$R(z^{-1}) = (1 + 2z^{-1} + z^{-2}) - (8 - 6z^{-1} + z^{-2}) = -7 + 8z^{-1} \quad (34)$$

Therefore:

$$H(z) = 8 + \frac{-7 + 8z^{-1}}{1 - 0.75z^{-1} + 0.125z^{-2}} \quad (35)$$

(c) Partial Fraction Expansion: Factor the denominator polynomial:

$$1 - 0.75z^{-1} + 0.125z^{-2} = (1 - 0.5z^{-1})(1 - 0.25z^{-1}) \quad (36)$$

Poles are $p_1 = 0.5$ and $p_2 = 0.25$. We expand the remainder strictly in powers of z^{-1} :

$$\frac{-7 + 8z^{-1}}{(1 - 0.5z^{-1})(1 - 0.25z^{-1})} = \frac{A_1}{1 - 0.5z^{-1}} + \frac{A_2}{1 - 0.25z^{-1}} \quad (37)$$

Evaluating residues via Heaviside's cover-up method:

$$A_1 = \left. \frac{-7 + 8z^{-1}}{1 - 0.25z^{-1}} \right|_{z^{-1} = \frac{1}{0.5} = 2} = \frac{-7 + 8(2)}{1 - 0.25(2)} = \frac{-7 + 16}{1 - 0.5} = \frac{9}{0.5} = \mathbf{18} \quad (38)$$

$$A_2 = \left. \frac{-7 + 8z^{-1}}{1 - 0.5z^{-1}} \right|_{z^{-1} = \frac{1}{0.25} = 4} = \frac{-7 + 8(4)}{1 - 0.5(4)} = \frac{-7 + 32}{1 - 2} = \frac{25}{-1} = \mathbf{-25} \quad (39)$$

Thus, the complete Parallel Form decomposition is:

$$\mathbf{H(z)} = \mathbf{8} + \frac{\mathbf{18}}{\mathbf{1 - 0.5z^{-1}}} - \frac{\mathbf{25}}{\mathbf{1 - 0.25z^{-1}}} \quad (40)$$

Verification Check ($z^{-1} = 0$):

$$H(0) = \frac{1 + 0 + 0}{1 - 0 + 0} = 1 \quad (41)$$

From (40):

$$8 + 18 - 25 = 1 \quad \checkmark \text{ (Exact match)} \quad (42)$$

(d) Difference Equations: Let $w_1[n]$ and $w_2[n]$ be the internal state variables for Branch 1 and Branch 2 respectively.

Branch 1 (Pole $p_1 = 0.5$, Residue $A_1 = 18$):

$$w_1[n] = x[n] + 0.5w_1[n - 1] \quad (43)$$

$$y_1[n] = 18w_1[n] \quad (44)$$

Branch 2 (Pole $p_2 = 0.25$, Residue $A_2 = -25$):

$$w_2[n] = x[n] + 0.25w_2[n - 1] \quad (45)$$

$$y_2[n] = -25w_2[n] \quad (46)$$

Total System Output:

$$\mathbf{y[n] = 8x[n] + y_1[n] + y_2[n] = 8x[n] + 18w_1[n] - 25w_2[n]} \quad (47)$$

Complete Parallel Realization for Example 19.1

$$H(z) = 8 + \frac{18}{1 - 0.5z^{-1}} - \frac{25}{1 - 0.25z^{-1}} \quad (M = 2, N = 2 \rightarrow C = 8)$$

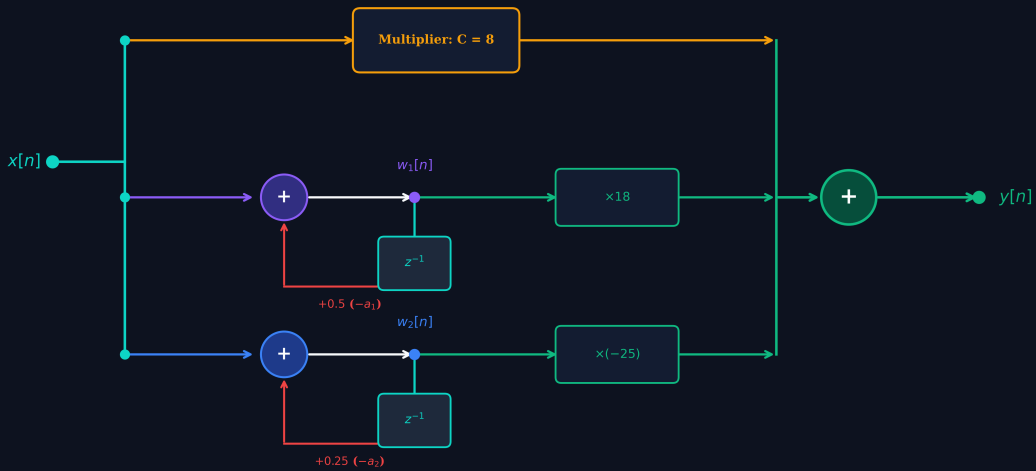


Figure 4: Complete Parallel Realization for Example 19.1. The input $x[n]$ feeds the direct feedthrough path ($C = 8$) and two first-order canonical branches ($p_1 = 0.5, p_2 = 0.25$). The branch outputs sum to yield $y[n]$.

(f) **Step-by-Step Impulse Response Evaluation** ($x[n] = \delta[n]$): Assuming initially relaxed registers ($w_1[-1] = 0, w_2[-1] = 0$):

- **At $n = 0$ ($x[0] = 1$):**

$$\begin{aligned} w_1[0] &= 1 + 0.5(0) = \mathbf{1.0} \implies y_1[0] = 18(1.0) = \mathbf{18.0} \\ w_2[0] &= 1 + 0.25(0) = \mathbf{1.0} \implies y_2[0] = -25(1.0) = \mathbf{-25.0} \\ y[0] &= 8x[0] + y_1[0] + y_2[0] = 8(1) + 18 - 25 = \mathbf{1.0} \end{aligned}$$

- **At $n = 1$ ($x[1] = 0$):**

$$\begin{aligned} w_1[1] &= 0 + 0.5(1.0) = \mathbf{0.5} \implies y_1[1] = 18(0.5) = \mathbf{9.0} \\ w_2[1] &= 0 + 0.25(1.0) = \mathbf{0.25} \implies y_2[1] = -25(0.25) = \mathbf{-6.25} \\ y[1] &= 8(0) + 9.0 - 6.25 = \mathbf{2.75} \end{aligned}$$

- **At $n = 2$ ($x[2] = 0$):**

$$\begin{aligned} w_1[2] &= 0 + 0.5(0.5) = \mathbf{0.25} \implies y_1[2] = 18(0.25) = \mathbf{4.5} \\ w_2[2] &= 0 + 0.25(0.25) = \mathbf{0.0625} \implies y_2[2] = -25(0.0625) = \mathbf{-1.5625} \\ y[2] &= 8(0) + 4.5 - 1.5625 = \mathbf{2.9375} \end{aligned}$$

The impulse response sequence is $h[n] = \{\mathbf{1.0}, \mathbf{2.75}, \mathbf{2.9375}, \dots\}$.

6.2 Example 19.2: Fourth-Order IIR System with Complex Conjugate Poles

Problem Statement: Realize the fourth-order strictly proper system in Parallel Form:

$$H(z) = \frac{4 - 3z^{-1} + 2z^{-2}}{(1 - 0.5z^{-1})(1 + 0.5z^{-1})(1 - \sqrt{2}z^{-1} + 0.64z^{-2})} \quad (48)$$

Solution:

1. **Classification:** Numerator degree $M = 2$, denominator degree $N = 4$. Since $M < N$, the system is strictly proper, so $\mathbf{C} = \mathbf{0}$.

2. **Pole Structure:**

- Real pole pair: $p_{1,2} = \pm 0.5 \implies (1 - 0.5z^{-1})(1 + 0.5z^{-1}) = 1 - 0.25z^{-2}$.
- Complex conjugate pole pair: $1 - \sqrt{2}z^{-1} + 0.64z^{-2}$. Comparing with $1 - 2r \cos(\theta)z^{-1} + r^2z^{-2}$: $r^2 = 0.64 \implies r = 0.8$. $2(0.8) \cos(\theta) = \sqrt{2} \implies \cos(\theta) = \frac{\sqrt{2}}{1.6} \approx 0.8839 \implies \theta \approx 27.89^\circ$.

3. **Decomposition into Two Second-Order Sections:**

$$H(z) = \frac{\beta_{01} + \beta_{11}z^{-1}}{1 - 0.25z^{-2}} + \frac{\beta_{02} + \beta_{12}z^{-1}}{1 - \sqrt{2}z^{-1} + 0.64z^{-2}} \quad (49)$$

Multiplying both sides by the denominator and equating coefficients of matching powers of z^{-1} yields the exact real numerator parameters:

$$H_1(z) = \frac{1.8286 - 0.7429z^{-1}}{1 - 0.25z^{-2}}, \quad H_2(z) = \frac{2.1714 - 1.1857z^{-1}}{1 - 1.4142z^{-1} + 0.64z^{-2}} \quad (50)$$

4. Hardware Resource Summary:

- Delays: $\max(1, 2) + \max(1, 2) = 2 + 2 = 4$ registers (strictly canonical, $N = 4$).
- Multipliers: 2 feedforward + 1 feedback (Section 1) + 2 feedforward + 2 feedback (Section 2) = **7** multipliers.
- Adders: $2 + 3 + 1 = 6$ adders.

6.3 Example 19.3: Reverse Synthesis — Extracting System Function from Parallel SFG

Problem Statement: A parallel digital filter realization has direct gain $C = 2$ and two first-order branches:

$$\begin{aligned}\text{Branch 1: } w_1[n] &= x[n] + 0.6w_1[n-1], & y_1[n] &= 3w_1[n] \\ \text{Branch 2: } w_2[n] &= x[n] - 0.4w_2[n-1], & y_2[n] &= -2w_2[n] \\ \text{Output: } y[n] &= 2x[n] + y_1[n] + y_2[n]\end{aligned}$$

Find the overall transfer function $H(z)$ and evaluate BIBO stability.

Solution: Transforming to the Z -domain:

$$H(z) = 2 + \frac{3}{1 - 0.6z^{-1}} - \frac{2}{1 + 0.4z^{-1}} \quad (51)$$

Combining over a common denominator:

$$\begin{aligned}H(z) &= \frac{2(1 - 0.6z^{-1})(1 + 0.4z^{-1}) + 3(1 + 0.4z^{-1}) - 2(1 - 0.6z^{-1})}{(1 - 0.6z^{-1})(1 + 0.4z^{-1})} \\ &= \frac{2(1 - 0.2z^{-1} - 0.24z^{-2}) + 3 + 1.2z^{-1} - 2 + 1.2z^{-1}}{1 - 0.2z^{-1} - 0.24z^{-2}} \\ &= \frac{2 - 0.4z^{-1} - 0.48z^{-2} + 1 + 2.4z^{-1}}{1 - 0.2z^{-1} - 0.24z^{-2}} \\ \mathbf{H(z)} &= \frac{\mathbf{3 + 2.0z^{-1} - 0.48z^{-2}}}{\mathbf{1 - 0.2z^{-1} - 0.24z^{-2}}} \quad (52)\end{aligned}$$

Stability Assessment: The system poles are $p_1 = 0.6$ and $p_2 = -0.4$. Since $|p_1| = 0.6 < 1$ and $|p_2| = 0.4 < 1$, all poles lie strictly inside the unit circle, confirming that the system is **BIBO stable**.

7. FACULTY INSIGHTS, COMMON MISCONCEPTIONS & EXAM TRAPS

1. Exam Trap 1: Forgetting the Direct Feedthrough Constant C when $M \geq N$:

- If a student attempts Heaviside cover-up expansion directly on an improper transfer function ($M = N$) without first dividing out C , the calculated residues will be ****completely wrong****!
- *Golden Rule:* Always check the highest powers of z^{-1} in numerator and denominator. If $M = N$, calculate $C = b_N/a_N$ first, subtract C , and expand only the strictly proper remainder.

2. Exam Trap 2: The Minus Sign in Branch Feedback Coefficients ($-a_{1k}, -a_{2k}$):

- In the branch denominator $1 + a_{1k}z^{-1} + a_{2k}z^{-2}$, if $a_{1k} = -0.75$, the state update is:

$$w_k[n] = x[n] - a_{1k}w_k[n-1] = x[n] - (-0.75)w_k[n-1] = x[n] + 0.75w_k[n-1]$$

The multiplier branch in the signal flow graph is labeled $+0.75$ (negated relative to the denominator coefficient).

3. Misconception 1: "Parallel Form can place transmission zeros independently in each branch."

- **Reality: False.** In Cascade Form, each biquad has its own zeros determined directly by $b_{k0} + b_{k1}z^{-1} + b_{k2}z^{-2}$. In Parallel Form, the zeros of $H(z)$ are created by the ****additive cross-cancellation (destructive interference)**** of all branches combined! Modifying a single branch alters all zero locations of the overall filter.

4. Misconception 2: "Parallel biquads need three feedforward multipliers like Cascade biquads."

- **Reality: Never.** Because partial fraction expansion of a proper second-order fraction always yields a linear numerator ($\beta_{0k} + \beta_{1k}z^{-1}$), there is no z^{-2} term in the feedforward path. Parallel biquads require only ****2 feedforward multipliers****, saving 1 multiplier per section.

8. SUMMARY & QUICK REVISION REFERENCE

Concept / Metric	Key Formula / Summary Rule
Parallel Transfer Function	$H(z) = C + \sum_{k=1}^K H_k(z) = C + \sum_{k=1}^K \frac{\beta_{0k} + \beta_{1k}z^{-1}}{1 + a_{1k}z^{-1} + a_{2k}z^{-2}}$
Direct Feedthrough Term (C)	$C = b_N/a_N$ if $M = N$; $C = 0$ if $M < N$ (strictly proper)
Branch State Equation	$w_k[n] = x[n] - a_{1k}w_k[n-1] - a_{2k}w_k[n-2]$
Branch Output Equation	$y_k[n] = \beta_{0k}w_k[n] + \beta_{1k}w_k[n-1]$
Total Output Equation	$y[n] = Cx[n] + \sum_{k=1}^K y_k[n]$
Complex Residue Grouping	$\beta_{0k} = 2\text{Re}(R_k), \beta_{1k} = -2\text{Re}(R_k p_k^*), a_{1k} = -2r_k \cos(\theta_k), a_{2k} = r_k^2$
Feedforward Multiplier Count	Strictly 2 per SOS (no $\beta_{2k}z^{-2}$ tap; saves 1 multiplier per biquad vs Cascade)
Quantization Noise Benefit	Zero inter-stage noise amplification ; all branch noise sources decouple independently
Critical Path Latency	Pipelined concurrent processing: 1 Mult + $(2 + \lceil \log_2 K \rceil)$ Adds

Table 2: Lecture 19 Quick Revision Reference.